AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129
Technologies in practice

Makeup-Midterm Examination (MTA) - Fall Semester 2024/2025

Date: 23/Apr./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 H

Instructions:

- Total Marks:**60**
- This exam consists of 4 Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [10 marks]: Choose The correct answer:

1. **Which of the following is a primary function of a router in a network?**
   a) To connect devices within the same LAN
   b) To forward data between different networks
   c) To provide wireless connectivity
   d) To manage network security
2. **What is the main function of the Data Link Layer in the OSI model?**
   a) Error detection and correction
   b) Logical addressing
   c) Data encryption
   d) Session management
3. **Which of the following is NOT a type of operating system?**
   a) Batch Operating System
   b) Time-Sharing Operating System
   c) Distributed Operating System
   d) Network Routing Operating System
4. **What is the purpose of a subnet mask in IP addressing?**
   a) To identify the network portion of an IP address
   b) To encrypt data packets
   c) To assign MAC addresses
   d) To manage DNS queries
5. **Which layer of the OSI model is responsible for data encryption and decryption?**
   a) Physical Layer
   b) Data Link Layer
   c) Presentation Layer
   d) Transport Layer
6. **What is the primary purpose of ARP (Address Resolution Protocol)?**
   a) To map IP addresses to MAC addresses
   b) To resolve domain names to IP addresses
   c) To assign IP addresses dynamically
   d) To encrypt network traffic
7. **Which of the following is a characteristic of a Time-Sharing Operating System?**
   a) Processes are executed in batches
   b) Multiple users can access the system simultaneously
   c) It is used only in embedded systems
   d) It does not support multitasking
8. **What is the role of the kernel in an operating system?**
   a) To manage system resources and act as an intermediary between software and hardware
   b) To provide a graphical user interface
   c) To manage network connections
   d) To encrypt data stored on disk
9. **Which of the following is a function of the Transport Layer in the OSI model?**
   a) Error detection and correction
   b) Logical addressing
   c) End-to-end communication and data flow control
   d) Data encryption
10. **What is the purpose of a default gateway in a network?**
    a) To connect devices within the same LAN
    b) To forward traffic from a local network to other networks
    c) To provide wireless connectivity
    d) To manage DNS queries

Question 2: Compare between the following [10 marks]:

1. Compare between TCP and UDP protocols. (2 Marks)
2. Compare between a hub and a switch in a network. (2 Marks)

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3. Compare between IPv4 and IPv6. (2 Marks)
4. Compare between preemptive and non-preemptive scheduling in operating systems. (2 Marks)
5. Compare between a collision domain and a broadcast domain. (2 Marks)

Question 3 [10 marks]

1. Match the networking protocols to their primary functions. (4 Marks)

Protocol	Function
1. TCP	a) Resolves domain names to IP addresses
2. UDP	b) Provides connection-oriented, reliable data transfer
3. DNS	c) Provides connectionless, faster data transfer
4. DHCP	d) Automatically assigns IP addresses to devices on a network

2. What is the role of the kernel in an operating system? How does it interact with hardware and user applications? (3 Marks)
3. What is the role of the dispatcher in CPU scheduling? (3 Marks)

Question 4 [10 marks]

1. Match the operating system concepts to their definitions. (4 Marks)

Concept	Definition
1. Virtual Memory	a) A situation where two or more processes are unable to proceed because each is waiting for the other to release resources
2. Deadlock	b) A technique that allows a system to use disk space as an extension of RAM
3. Paging	c) Divides memory into fixed-size blocks called pages
4. Interrupt	d) A signal to the CPU to pause its current task and handle a high-priority event

2. Explain the concept of deadlock in operating systems. What are the necessary conditions for a deadlock to occur, and how can it be prevented or resolved? (3 Marks)
3. What is the purpose of the DNS (Domain Name System) in networking? How does it work, and why is it critical for internet communication? (3 Marks)

Question 5 [10 marks]

1. What is the purpose of a linker and a loader in an OS? (3 Marks)
2. What is resource allocation in OS services? (3 Marks).
3. Contrast the security capabilities of WPA2 and WEP protocols in securing wireless networks. What are the advantages and limitations of each? (4 Marks)

Question 6 [10 marks]

1. Explain the concept of virtualization in operating systems. How does it improve resource utilization, and what are some common use cases for virtualization? (3 Marks)
2. Describe the process of context switching in operating systems. Why is it necessary, and what are the potential performance impacts? (3 Marks)
3. What is the difference between symmetric multiprocessing (SMP) and asymmetric multiprocessing (AMP)? Provide an example of a scenario where each might be used. (4 Marks)

End of Questions

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